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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT



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(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

“Tourism Data Analysis Using Power BI”

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI “**Tourism Data Analysis Using Power BI**” has been successfully presented by **S Gayathri** bearing USN **3BR23CD080** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

Mr. Sundeep Sindhanoor

Signature of HOD

Dr. Aradhana D

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1: Introduction

1.1 About the Project: Tourism is one of the fastest-growing industries and plays a major role in the economic development of a country. Understanding tourist behavior, seasonal trends, and revenue generation is essential for improving tourism services and planning strategies.

1.2 The purpose of this project is to analyze tourism data using a structured dataset containing information such as number of visitors, location, season, revenue, and tourist preferences. The dataset is imported into Power BI, where data cleaning, transformation, and visualization techniques are applied to extract meaningful insights.

1.3 The final outcome of this project is an interactive dashboard that highlights tourist trends, seasonal variations, popular destinations, and revenue analysis. This dashboard helps stakeholders understand tourism patterns and make better decisions for improving tourism services.

1.4 Language Used

The project was implemented using Microsoft Power BI.

- **Dataset (Excel/CSV):** Tourism data including visitors, season, revenue, and locations.
- **Power Query:** Used for data cleaning and transformation such as removing duplicates and handling missing values.
- **DAX (Data Analysis Expressions):** Used to calculate KPIs like total visitors, revenue, and growth rate.
- **Power BI Visualization Tools:** Used to create bar charts, line charts, pie charts, and KPI cards

By combining CSV data with Power BI's powerful features, the project successfully delivered a clear, interactive dashboard that explains transport reliability in a simple and effective way.

2: Scope of the Project

Problem Statement: Tourism industries generate large amounts of data, but analyzing this data manually is difficult and time-consuming. Without proper visualization, it is hard to identify trends such as peak seasons, most visited locations, and revenue patterns.

Traditional methods fail to provide interactive insights, making it difficult for decision-makers to plan tourism strategies effectively. Hence, there is a need for a data-driven approach using visualization tools like Power BI to convert raw tourism data into meaningful insights.

2.2 Objectives:

- Import and clean tourism dataset using Power BI.
- Analyze tourist visits based on seasons and locations.
- Calculate KPIs such as total visitors and revenue.
- Create interactive dashboards for better understanding.
- Identify trends and patterns in tourism data.

3: System Requirements

3.1 Functional Requirements

- Import tourism dataset from Excel/CSV files into Power BI without errors
- Clean and transform data using Power Query by handling missing values, removing duplicates, and formatting columns
- Create calculated columns such as total visitors per location and seasonal categorization
- Develop DAX measures to calculate key metrics such as total visitors, total revenue, average visitors, and growth trends
- Build interactive visualizations such as bar charts, column charts, line charts, pie charts, and KPI cards

Non-Functional Requirements Non-functional requirements ensure the system is efficient, reliable, and easy to use.

- **Usability:** Dashboard must be simple and intuitive for non-technical users.
- **Performance:** Visualizations should load quickly even with large datasets.
- **Scalability:** System should handle additional datasets if more transport data is added in the future.
- **Accuracy:** Calculations of KPIs and delay minutes must be precise.

3.2 Hardware Requirements To run Power BI effectively and handle CSV datasets, the following hardware is recommended:

- A computer or laptop with at least **8 GB RAM**.
- Processor: Intel i5 or equivalent (minimum dual-core).
- Storage: At least 500 GB HDD/SSD for dataset storage and software installation.
- Display: Full HD resolution (1920x1080) for clear visualization of dashboards.

3.3 Software Requirements The software environment required for the project includes:

- **Microsoft Power BI Desktop** (for data import, transformation, and visualization).
- **Power Query** (integrated within Power BI for data cleaning and preparation).
- **DAX (Data Analysis Expressions)** (for creating measures and calculations).
- **CSV Dataset File** (raw transport data containing trip details, vehicle type, delay cause, and timings).

4: Methodology

4.1 Data Collection The data collection process is the foundation of this project, as the quality of analysis depends on the quality of data used. In this project, the tourism dataset was collected in a structured format containing important attributes such as tourist locations, number of visitors, seasons, and revenue generated. The dataset represents realistic tourism trends and provides sufficient information for meaningful analysis.

The data was stored in Excel/CSV format, which makes it easy to import into Power BI without requiring complex database connections. The dataset includes records from different time periods, allowing the analysis of seasonal variations and growth trends over time. Care was taken to ensure that the data is consistent and relevant to the objectives of the project.

4.2 Data Preparation (Power Query) After collecting the dataset, the next step is data preparation, which is performed using Power Query in Power BI. This step is essential to ensure that the data is clean, accurate, and suitable for analysis. Raw data often contains errors such as missing values, duplicate entries, and inconsistent formats, which must be corrected before visualization.

In this project, Power Query was used to clean and transform the dataset. Missing values were identified and either filled or removed depending on their importance. Duplicate records were eliminated to avoid incorrect analysis. Column names were standardized, and data types were assigned correctly, such as converting date fields into proper date format and numerical values into appropriate data types.

4.3 Data Visualization Visualization was the core of the methodology, as the project aimed to transform raw tourism data into meaningful insights. Power BI's visualization tools were used to create a variety of charts and KPI cards that clearly represent tourism trends and patterns.

- **KPI Cards** displayed key metrics such as total number of visitors, total revenue generated, and average visitors per location.
- **Bar Charts** compared the number of visitors across different tourist locations, helping to identify the most popular destinations.

- **Line Charts** illustrated tourism trends over time, showing how the number of visitors changes across different seasons or years.
- **Pie Charts** showed the distribution of visitors across various locations or seasons, highlighting which category has the highest contribution.
- **Revenue Analysis Charts** explained how revenue varies by location and season, helping to understand which areas generate the highest income.

4.4 Filters and Interactivity Filters and interactivity are important features of this project, as they allow users to explore tourism data dynamically instead of viewing it as a static report. By using Power BI's slicers and filters, the dashboard becomes more flexible and user-friendly, enabling users to focus on specific aspects of the data.

In this project, different types of filters were applied to enhance analysis:

Location Filter: Allows users to view data for a specific tourist destination and compare its performance with others.

Season Filter: Helps in analyzing tourism trends across different seasons such as summer, winter, and monsoon.

Year/Date Filter: Enables users to observe changes in tourism patterns over time.

Revenue Filter: Helps in focusing on high or low revenue-generating locations.

4.5 Dashboard Design Filters and interactivity are important features of this project, as they allow users to explore tourism data dynamically instead of viewing it as a static report. By using Power BI's slicers and filters, the dashboard becomes more flexible and user-friendly, enabling users to focus on specific aspects of the data.

- KPI cards are placed at the top to provide a quick summary of important metrics such as total visitors and revenue.
- Charts such as bar graphs and line charts are placed in the middle to show detailed comparisons and trends.
- Pie charts and distribution visuals are included to highlight proportions and contributions.
- Filters and slicers are positioned on the side or top to allow easy interaction.

Outcome The final outcome of this project is an interactive Power BI dashboard that provides a comprehensive analysis of tourism data. The dashboard successfully transforms raw data into meaningful insights, making it easier to understand tourism trends and patterns.

The analysis revealed key findings such as the most visited tourist locations, peak seasons with the highest number of visitors, and revenue distribution across different destinations. It also helped in identifying patterns such as seasonal fluctuations in tourism and variations in revenue generation.

Overall, the dashboard serves as a powerful tool for decision-making. It can help tourism authorities and planners improve strategies, allocate resources effectively, and enhance tourist experiences. The project demonstrates how data visualization can be used to gain valuable insights from tourism data.

5: Implementation

5.1 Dataset Description The dataset used in this project was provided in Excel/CSV format and contains tourism-related information. It includes important attributes such as tourist locations, number of visitors, season, and revenue generated. These fields are essential for analyzing tourism trends and understanding visitor behavior.

5.2 Data Loading The dataset was imported directly into Microsoft Power BI Desktop using its built-in data import feature. Power BI supports Excel and CSV formats, which made the loading process simple and efficient.

Once the data was loaded, it was displayed in the data view, where all columns and records could be verified. This step ensured that the dataset was correctly imported and ready for further processing. Proper data loading is important to avoid errors in later stages of analysis.

5.3 Data Transformation (Power Query) After loading the dataset, data transformation was carried out using Power Query. This step involved cleaning and preparing the data to ensure accuracy and consistency. Raw data often contains errors, so it must be processed before analysis.

5.4 Data Modelling Data modelling is the process of organizing the dataset into a structured format that supports efficient analysis. In this project, the dataset was arranged in a logical way so that relationships between data fields could be used effectively.

If multiple tables are used, relationships can be created between them based on common fields such as location or date. Even with a single dataset, proper structuring ensures that calculations and visualizations work correctly. Data modelling improves performance and accuracy of the dashboard.

5.5 DAX Measures (Key Calculations) DAX (Data Analysis Expressions) was used to create important measures that drive the insights in the dashboard. These measures help in summarizing and analyzing the data effectively.

Some key calculations used in this project include:

- **Total Visitors:** Calculates the total number of tourists across all locations
- **Total Revenue:** Calculates the total income generated from tourism
- **Average Visitors:** Finds the average number of visitors per location
- **Seasonal Growth:** Measures changes in visitors across different seasons

These measures provided the quantitative backbone of the dashboard

5.6 Dashboard Visualization Power BI's visualization tools were used to create various charts and graphs that represent tourism data. KPI cards display key metrics such as total visitors and revenue, while bar charts compare different locations. Line charts show trends over time, and pie charts represent the distribution of visitors.

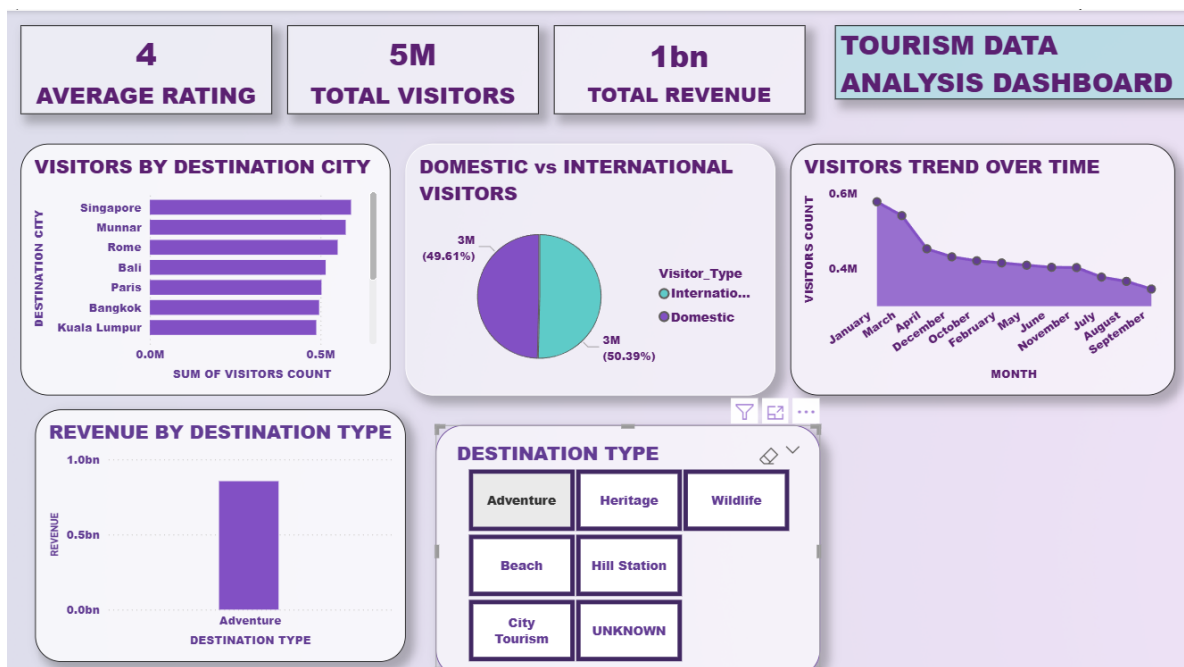
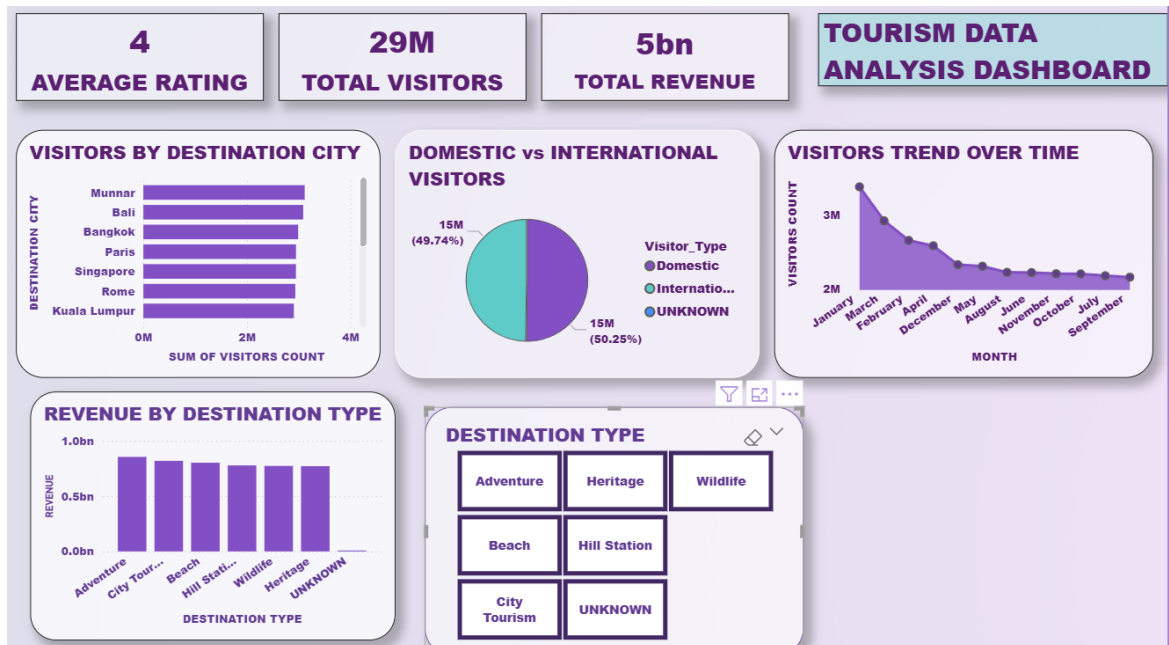
5.7 Filters / Slicers Filters and slicers play a crucial role in enhancing the interactivity of the dashboard by allowing users to explore tourism data dynamically. Instead of viewing static information, users can apply filters to focus on specific aspects such as location, season, or time period. In this project, slicers were added for tourist locations, seasons, and year/date, enabling users to analyze patterns in a more detailed manner. For example, users can select a particular location to view its visitor count and revenue or choose a specific season to understand seasonal trends. Additional filters such as revenue range and visitor count further help in identifying high-performing and low-performing destinations. These interactive features allow users to drill down into the data and customize their analysis according to their needs, making the dashboard more flexible, informative, and user-friendly.

5.8 Dashboard Layout The dashboard layout was designed in a clear and organized manner. KPI cards were placed at the top to provide a quick overview, followed by charts and graphs that explain the data in detail. Filters were positioned for easy access, ensuring smooth interaction.

The layout ensures that users can easily navigate the dashboard and understand the insights..

5.9 Final Output The final output of this project is an interactive Power BI dashboard that provides a comprehensive analysis of tourism data in a clear and meaningful way. The dashboard integrates all the processed data, visualizations, and filters into a single interface, allowing users to explore tourism trends effectively. It highlights key insights such as the most visited locations, seasonal variations in tourist arrivals, and revenue distribution across different destinations. By using interactive features, users can customize their analysis and focus on specific data points, which enhances the overall understanding of tourism patterns. The dashboard serves as a valuable tool for decision-making, helping tourism planners and stakeholders identify opportunities, improve strategies, and allocate resources efficiently. Overall, the project demonstrates how raw tourism data can be transformed into actionable insights .

Result



Conclusion

The “Tourism Data Analysis using Power BI” project successfully demonstrates how raw tourism data can be transformed into meaningful and actionable insights through the use of data visualization techniques. By systematically collecting, cleaning, and preparing the dataset using Power Query, and applying DAX measures for key calculations, the project was able to analyze important aspects such as visitor trends, seasonal variations, and revenue generation across different tourist locations. The interactive dashboard developed in Power BI serves as the central outcome of this project, providing a clear and comprehensive view of tourism patterns. It enables users to explore data dynamically through filters and slicers, making it easier to identify popular destinations, peak seasons, and areas that require improvement.

The analysis highlights how certain locations attract more visitors and generate higher revenue, while others may need better promotion or infrastructure development. Seasonal trends observed in the dashboard also provide valuable insights for planning tourism activities and managing resources efficiently during peak and off-peak periods. The project not only focuses on identifying patterns but also emphasizes the importance of data-driven decision-making in the tourism sector. By using this dashboard, tourism authorities and stakeholders can make informed decisions, improve strategies, and enhance the overall tourist experience.

In conclusion, this project proves that Power BI is a powerful tool for analyzing and visualizing tourism data. It shows that even a simple dataset, when processed and presented effectively, can provide deep insights and support strategic planning. The methodology and approach used in this project can be extended to other domains where data analysis and visualization play a critical role. Overall, the project highlights the importance of transforming data into knowledge and using it to drive better decisions in the tourism industry.

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